



How do urban park features affect cultural ecosystem services: Quantified evidence for design practices

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ABSTRACT

Cultural ecosystem services (CES) in urban parks are associated with nature contact willingness and health outcomes of urban residents. However, practical knowledge on how to manage urban park features to enhance CES is still lacking, especially on more detailed scales. This study developed a practice-oriented workflow for exploring design-related indicators that affect CES by considering four aspects of urban park features, including distance to elements, density of elements, land cover proportion, and landscape diversity. Then the workflow was implemented in a case study by taking six urban parks in Beijing as study areas. Three CES types were identified with outdoor social media images, including recreation value, aesthetic value, and social interaction value. Statistical analyses indicated that a combination of the four aspects of urban park features within a specific service radius was the most significant factor in explaining CES. The importance of design-related indicators for enhancing different CES types was also identified. Density of facilities, proportion of tree canopy-shaded ground, and richness of land cover types were proved important for all the three CES types and therefore could be concerned in urban park design practices. Based on these findings, this study further proposed landscape site design strategies and three adaptive design cases for enhancing different CES types, which could provide evidence-based and practical solutions for managers and landscape architects.

1. Introduction

Ecosystem services have been increasingly integrated into urban environmental management and decision-making as a concept to help us understand and value the benefits of ecosystems to human well-being (de Groot et al., 2010). Urban ecosystems provide multiple ecosystem services for residents and can increase urban resilience (Gómez-Baggethun and Barton, 2013). Among them, cultural ecosystem services (CES) are the non-material benefits that people can obtain from urban ecosystems, which could potentially motivate people's willingness to have contact with nature and improve their health outcomes (MEA, 2005; Clark et al., 2014; Dickinson and Hobbs, 2017). With rapid urbanization and the travel restrictions due to the COVID-19 pandemic since 2020, urban parks and other open green spaces have become the main locations for human-nature interaction and the most important spatial carrier for the supply of CES (Kabisch et al., 2015; Zhang et al., 2022). The importance of conserving and enhancing CES through efficient urban park design practices has been highlighted for citizens' stress reduction and quality of life in this unprecedented time

(Beckmann-Wübbelt et al., 2021; Venter et al., 2021; Addas and Maghrabi, 2022). However, strategies for managing CES with urban park design practices are still relatively unsupported by evidence because spatially identifying and attributing CES on more detailed scales remains challenging (Schröter et al., 2015; Havinga et al., 2020).

CES emphasizes experiential interactions with nature, and its assessment was consequently focused on participatory methods (Schirpke et al., 2016; van Riper et al., 2017), including interviews, social surveys, and public-participation GIS (PPGIS) approaches (Plie-ninger et al., 2013; Bryce et al., 2016; Komossa et al., 2020; Gai et al., 2022), which are time- and resource-consuming. Therefore, geo-referenced social media data in the forms of text, images, and videos have gradually been applied and transformed as a proxy for perceived CES (Havinga et al., 2020; Heikinheimo et al., 2020; Lee et al., 2020). The density of geo-tagged social media images provides a means of mapping the distribution of CES and assessing changes over space and time (Wood et al., 2013; Richards and Friess, 2015). Recently, rapidly improving computer vision approaches have enabled automated content identification of images (Voulodimos et al., 2018; Yao et al., 2019),

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which could provide a large amount of information that is broadly relevant to people's contacts with the natural environment (Seiferling et al., 2017) and represents environmental features that are attractive to the public (Di Minin et al., 2015). This information can be used to distinguish different CES types (Richards and Tunçer, 2018). Such studies have demonstrated that social media data can be valuable for understanding and mapping CES over large areas (Cardoso et al., 2022; Crouzat et al., 2022; Lee et al., 2022), but may be limited when assessing CES on more detailed scales. This gap is mainly caused by the limited spatial reference accuracy of traditional social media (analysis units are generally grids larger than $2 \text{ km} \times 2 \text{ km}$), which place particular emphasis on online social functions. However, in urban areas, many human-nature contacts are generated by outdoor activities, which mostly occur in small and heterogeneous urban green spaces (McCormack et al., 2010). Unlike common social media, outdoor social media are dedicated to providing a precise positioning service for users and can support sharing of images taken at specific locations. Therefore, although data from outdoor social media have not been given wide attention, they may provide opportunities for assessing CES in urban parks.

Linking spatial CES data with environmental features may help to understand the attribution of CES and generate information for managing CES in planning and design practice (Baumeister et al., 2020). Previous studies mainly hypothesized that such associations were related to specific characteristics (Brown, 2013) such as different land covers, degrees of vegetation coverage and landscape openness, and presence of natural and cultural elements (Hong and Guo, 2017; Vigl et al., 2017; Schirpke et al., 2018; Andreeva, 2019; Crouzat et al., 2022). By considering the role of various environmental features and their effects on CES, recent studies have conducted spatial modeling analyses based on maximum entropy (MaxEnt) modeling and Social Values for Ecosystem Services (SolVES) (Sherrouse et al., 2022) with more indicators such as elevation, slope, distance to roads, distance to water, land cover, and landforms. The importance of distance-related indicators in explaining CES was specifically emphasized in these literature (Richards and Tunçer, 2018; Gerstenberg et al., 2020). However, most of these indicators were designed to describe spatial heterogeneity on relatively large scales, which makes them impractical to a certain extent for design and management practice in urban parks. Indicators including density of roads and facilities, and landscape diversity in

specific service ranges are more direct indicators for guiding the park design process and creating landscape sites in urban parks (Golčnik and Ward Thompson, 2010), but the role of these aspects in supporting CES remains undiscovered.

To date, researches on CES in urban parks have generally ended with assessment (Johnson et al., 2019; Tian et al., 2021). Limited attention has been given to identifying the effect of urban park features on CES in high-density cities compared to natural ecosystems (Baumeister et al., 2020; Gerstenberg et al., 2020; Havinga et al., 2020). Given this insufficiency of evidence for practical design strategies to enhance CES in urban parks, this study will attempt to identify CES in urban parks with outdoor social media images and computer vision approach on a more detailed scale, and to explore design-related urban park features that affect CES. To fulfill the aim of this study, three research questions were formulated: 1) how to identify different CES types in urban parks at a more detailed scale? 2) how do urban park features contribute to explaining CES? and 3) how to enhance CES through urban park design practices? In the following sections, six urban parks in Beijing, China were chosen as a case study area. The results were expected to provide information for managers and landscape architects to enhance CES through urban park design practices that can promote outdoor activities and health outcomes of urban residents.

2. Methods

As shown in Fig. 1, a practice-oriented workflow was developed for the case study. The research questions will be addressed through the following processes. First, based on a deep learning-based computer vision approach and questionnaire survey, different CES types in urban parks will be identified with outdoor social media images. Second, we summarized four aspects of urban park features based on reviewing CES supply-related urban park features in previous literature (Golčnik and Ward Thompson, 2010; Gai et al., 2022; Zhang et al., 2022), including distance to elements, density of elements, land cover proportion, and landscape diversity. These urban park features will be mapped on fine spatial resolution, and the last three aspects will be calculated within different service radii. According to the findings of current studies on interactions between visitors and urban park environments (Graves et al., 2017; Gerstenberg et al., 2020; Havinga et al., 2020; Wang et al., 2021; Fu et al., 2022), this study adopted 10, 30, 50, and 100 m as

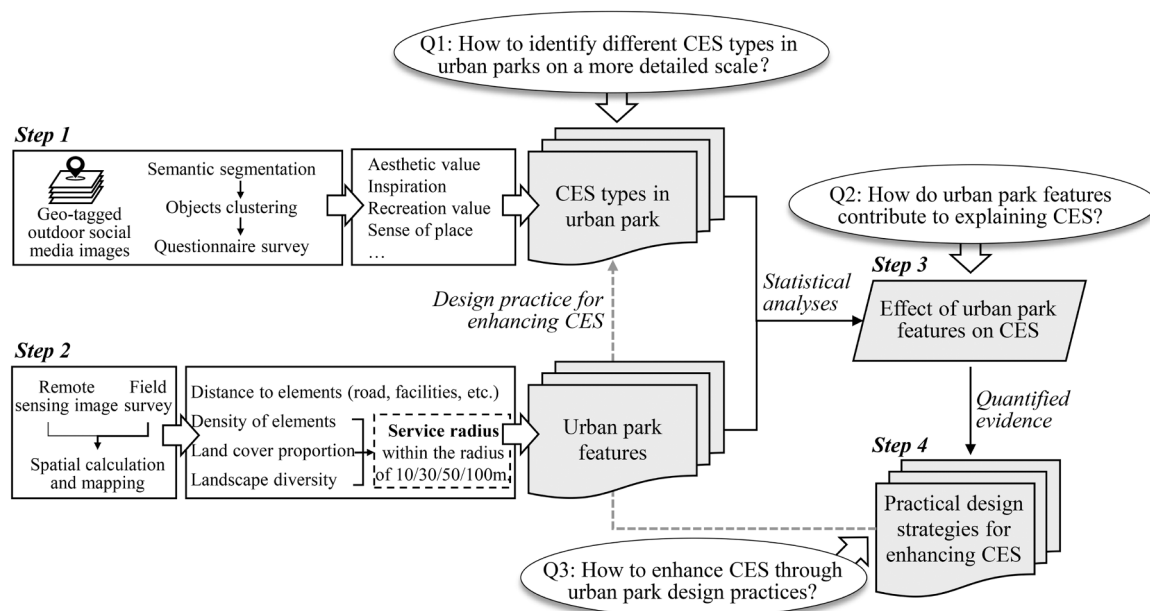


Fig. 1. Practice-oriented workflow for the case study.

service radii for urban park features. Then, the effect of urban park features on CES will be identified based on structural equation model (SEM) and multiple regression analyses. Finally, practical design strategies for enhancing CES in urban parks will be proposed based on quantified evidence from analysis results, which could be helpful for urban park design practices.

2.1. Study area

As a famous historical and tourist city, Beijing has plenty of historical parks and tourist attractions, such as the Summer Palace and the Olympic Forestry Park. In this study, to identify the CES of urban parks for daily outdoor activities, six urban parks that are more popular with local residents than with foreign tourists were selected as case studies: Chaoyang Park (CY), Zizhuyuan Park (ZY), Haidian Park (HD), Diaosu Park (DS), Longtan Park (LT), and Zhongshan Park (ZS) (Fig. 2). The selected parks serve all age groups and are widely distributed in the four main districts of Beijing (Haidian, Chaoyang, Dongcheng, and Shijingshan) with areas from 33 ha to 400 ha (the area of each park is indicated in Fig. 2).

2.2. Identifying CES with outdoor social media images and computer vision

2.2.1. Collection and semantic segmentation of outdoor social media images

Sixfoot is one of the most popular social application programs for outdoor activities in China. Released in 2010, Sixfoot provides services both on mobile phones and online. Users can share their travel itineraries and photos publicly with geographic coordinates. These records can be viewed by others on mobile terminals and downloaded freely on the Web (www.fooooooot.com). This study downloaded records that

were entirely took place in the surveyed urban parks, and extracted information about the images and their locations. Repeated uploaded records and those taken during nighttime (before 6:00 and after 18:00) were removed to increase accuracy. The process finally collected 755 records with 7291 images in the six urban parks from Sixfoot.

A fully convolutional network (FCN) model trained by the ADE-20 K dataset was used to semantically segment and extract visual objects from the outdoor social media images. This model has been applied to outdoor photographs of Chinese cities with estimation accuracy exceeding 90% on average (Yao et al., 2019; Sun and Lu, 2022). The segmented visual objects were classified into eight categories, with seven categories of objects inside urban parks and one category of other objects (Table 1).

2.2.2. Clustering and identifying different CES types

The area proportion of the eight categories of objects in each image were then calculated, and images containing more than 60% of other objects were excluded. Principal component analysis and *k*-means cluster analysis were applied to identify image groups with similar object formation characteristics. Then a randomly selected subset of 10 images from each cluster were shown to participants, who classified them subjectively into one of the ten given CES types according to their experience and perception (MEA, 2005). The questionnaire survey was conducted online in Chinese, and the translated questionnaire is provided in the Supplementary Materials (Table S1). A total of 60 respondents participated in the questionnaire survey, including 20 experts (professors and practitioners of urban ecology or landscape design fields), 20 graduate students (major in landscape architecture), and 20 non-professionals. The voted CES types in the questionnaire survey were linked to the locations of the analyzed outdoor social media images, and the density of image locations was mapped to represent the intensity of CES (Gliozzo et al., 2016; Tenerelli et al., 2016). The kernel density tool in ArcGIS 10.2 was used to process location point data into CES intensity

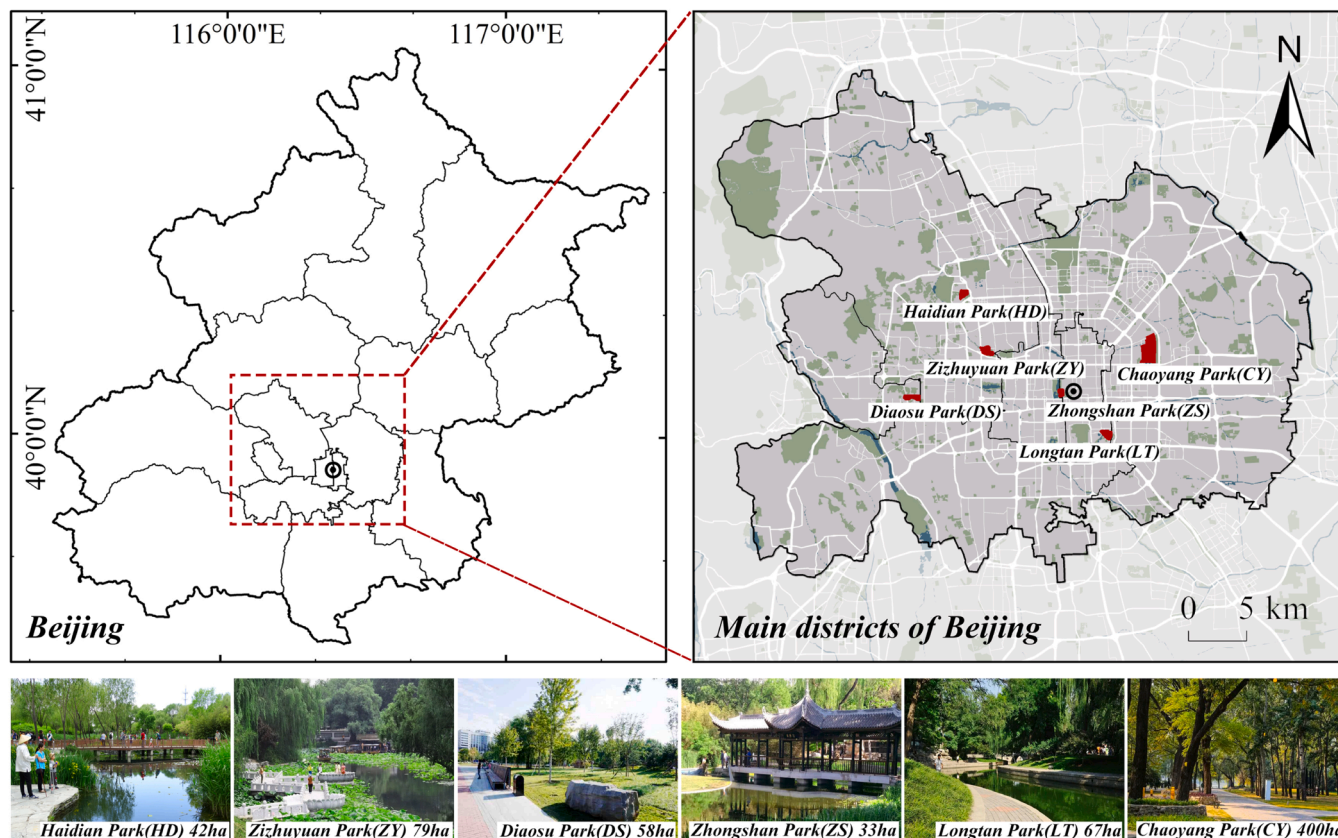


Fig. 2. Location of the six urban parks under analysis in Beijing.

Table 1
Image segmentation workflow and categories of segmented objects.

Workflow	Categories	Segmented objects
	Objects	Tree
	inside	Grass
	urban	Flower
	parks	Water
	Artificial objects	Impervious surfaces
Other objects	Facility	Road, pavement, path, ground, field.
	Human	Pillar, bench, sculpture, rockery, etc.
		Unknown objects; sky, animal, car, food, etc.

values at a spatial resolution of 1×1 m (Gerstenberg et al., 2020; Havinga et al., 2020), and the search radius was set to 50 m to represent the perceived surroundings of people in urban parks (Baumeister et al., 2020; Gerstenberg et al., 2020). To assess the spatial differentiation and spatial autocorrelation of the various CES types mapped in the surveyed urban parks, the Kruskal-Wallis one-way ANOVA test and the global Moran's I were used to analyze their spatial patterns (Reilly et al., 2018).

2.3. Mapping urban park features

To comprehensively understand urban park features and their effect on CES and to explore the most effective site solutions of plants, paving, and facility layout in urban park design practice, two hypotheses were proposed: 1) considering more aspects of park features could perform better in explaining CES, and 2) urban park features may perform differently in explaining CES within various service radii.

Therefore, a total of 20 design-related indicators were proposed to describe urban park features (Table 2). Among the 20 design-related indicators, water, roads, park squares, entrances, and facilities were included as urban park elements. Facilities here were further recorded as ornamental facilities (such as sculptures, fountains, and rockery), resting facilities (such as benches, pavilions, and chess tables), and interactive facilities (such as children's play facilities, swings, fitness equipment, and leisure equipment) according to their functions. Land cover types in urban park included impervious ground, water, canopy-shaded ground, evergreen trees, deciduous trees, shrubs, and grass. Data for calculating these indicators were obtained from field investigation and remote-sensing image processing.

The field investigation was conducted from September 20 to October 10, 2021. First, drawings of each park design were collected online and digitized to extract the spatial distributions of park roads, water bodies, and squares. Then these distributions were supplemented and corrected based on a handheld GPS (UniStrong, MG758) during the field investigation, and facility locations and park entrances were also marked and recorded as points. Finally, the field survey data were processed in ArcGIS 10.2 to obtain a digital map of the distributions of elements in the surveyed urban parks.

Cloud-free GF-2 satellite images with a high spatial resolution of 1×1 m and four spectral bands were used to identify different land cover types. The normalized difference vegetation index (NDVI) from GF-2 satellite image obtained in summer (June 6, 2017) were first calculated to identify vegetation areas, impervious surfaces, and water bodies. According to the image texture characteristics and the digital map of the parks, vegetation areas were further classified into four plant categories: canopy-shaded ground, trees, shrubs, and grass. The tree area was further divided into deciduous and evergreen trees by processing GF-2 satellite images obtained in winter (February 22, 2018) and identifying the trees with green leaves in winter. Land cover accuracy

Table 2
Urban park features and design-related indicators.

Urban park feature	Design-related indicator and abbreviation	Service radius
Distance to elements ($DIST_N$)	Distance to water (m)	$DIST_W$
	Distance to roads (m)	$DIST_R$
	Distance to park squares (m)	$DIST_S$
	Distance to entrances (m)	$DIST_E$
	Distance to facilities (m)	$DIST_{FO}$
	Ornamental facilities	$DIST_{FR}$
	Resting facilities	$DIST_{FI}$
Density of elements ($DENS_N_A$)	Density of roads (m/m^2)	$DENS_{R_A}$
	Density of facilities (per m^2)	$DENS_{FO_A}$
	Ornamental facilities	$DENS_{FR_A}$
	Resting facilities	$DENS_{FI_A}$
	Interactive facilities	$DENS_{FI_A}$
Land cover proportion ($LAND_N_A$)	Land cover proportion (%)	$LAND_{I_A}$
	Impervious ground	$LAND_{W_A}$
	Water	$LAND_{C_A}$
	Canopy-shaded ground	$LAND_{E_A}$
	Evergreen trees	$LAND_{D_A}$
	Deciduous trees	$LAND_{S_A}$
	Shrubs	$LAND_{G_A}$
	Grass	$LAND_{G_A}$
Landscape diversity ($DIVE_N_A$)	Shannon-Wiener index of land cover types	$DIVE_{S_A}$
	Richness index of land cover types	$DIVE_{R_A}$

was verified by checking 100 random points. The ground truth of the land cover at each sampling point was obtained from both field investigation and Google Earth images. The overall accuracy was 92%, and the accuracy of the vegetation areas was 88%, which met the accuracy requirements for estimation.

Fig. 3 shows the land cover areas and the distribution of elements in the surveyed urban parks. All urban park features were visualized at a spatial resolution of 1×1 m, and detailed calculation methods and maps for each indicator are presented in the Supplementary Materials (Tables S2–S5).

Land covers and distribution of elements

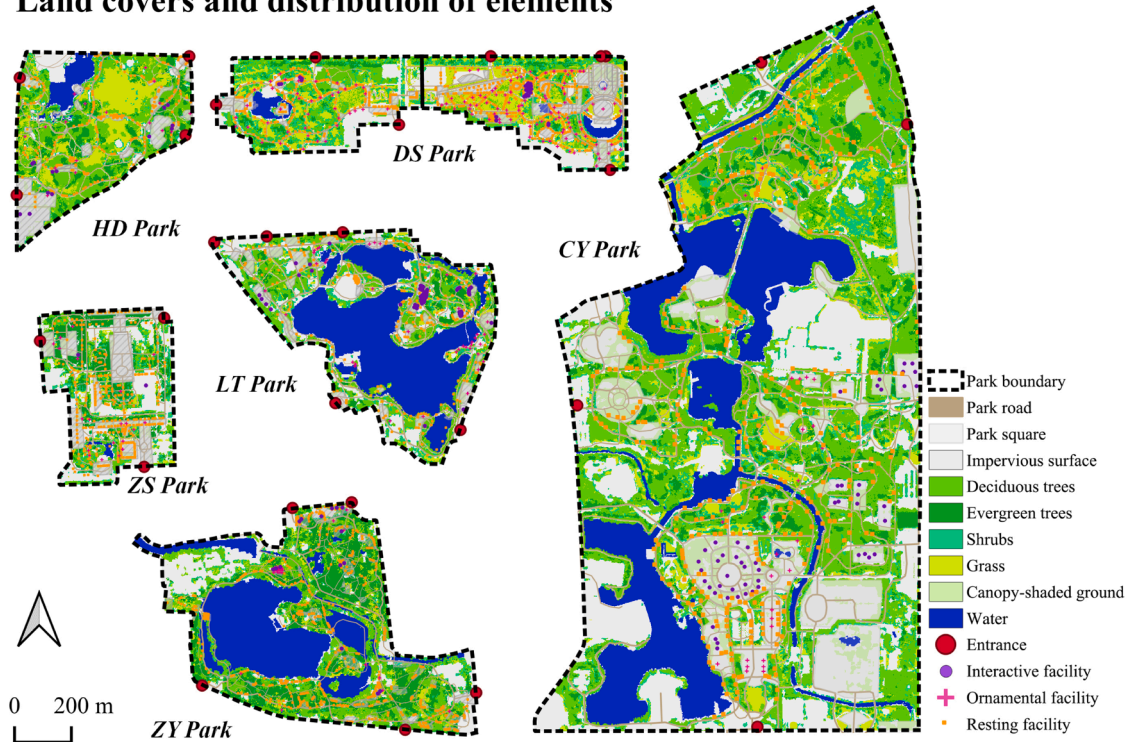


Fig. 3. Land cover and distribution of elements in surveyed urban parks.

2.4. Statistical analysis for explaining CES

2.4.1. Examining the performance of the hypothesis models

To examine the influence of urban park features on CES and identify the most significant service radius for each aspect, the hypothesis models illustrated in Fig. 4 were proposed. By taking CES intensity as the dependent variable, Model 1 served as a baseline model that considered only land cover proportion (LAND) and distance to elements (DIST) as predictors. These two elements have been widely proved to have significant influence on CES. Model 2 and Model 3 respectively added the predictors of density of elements (DENS) and landscape diversity (DIVE), and Model 4 contained predictors of both density of elements (DENS) and landscape diversity (DIVE).

Compared to the baseline model, the last three models attempted to examine the hypothesis that additional urban park features could optimize the performance of prediction models. Each service radius of an urban park feature was estimated in the hypothesis models to examine whether urban park features would perform differently in explaining CES depending on the service radius. CES and urban park features were standardized and extracted from raster data before analysis. A series of SEM analyses were then performed to determine model performance, with the goodness of fit represented by *R*-squared. The Analysis of Moment Structures software (Amos 23.0) was used to initialize the path diagrams and identify the modeling results (Gallagher et al., 2008). Indicators with a variance inflation factor (VIF) larger than the problematic value of 5 were excluded to minimize multicollinearity.

In Fig. 4, CES_i is the intensity value of the *i*-th CES type, which was the dependent variable of the model, including the voted recreation value, aesthetic value, and social interaction value. To achieve a linear relationship between variables, the CES intensity value was logarithmically transformed. $LAND_N_A$, $DIST_N$, $DENS_N_A$, and $DIVE_N_A$ were predictors. Specifically, based on preliminary analysis, $LAND_N_a$ was assigned as the most significant service radius urban park elements after analysis of Model 1, and $DENS_N_b$ and $DIVE_N_c$ were the most significant service radii of urban park elements in Model 2 and Model 3

respectively.

2.4.2. Multiple regression analyses for predicting CES

According to the SEM analysis, urban park features within specific service radii in the most significant hypothetical model were included in the multiple regression analyses to predict CES. Taking corresponding design-related indicators as independent variables, a series of multiple stepwise regression analyses were conducted to identify their importance in explaining different CES types. The regression coefficients (β) represent the importance of various urban park features in explaining CES, which would be valuable for providing evidence-based urban park design strategies. Residuals of the regression model for each CES type were normally distributed in this study. The processing and analyses were conducted in R 4.1.1 (R Core Team, 2021).

3. Results

3.1. CES in surveyed urban parks

3.1.1. CES types in surveyed urban parks

According to the object categories identified by semantic segmentation of outdoor social media images, 472 images containing more than 60% of other objects were excluded. Principal component analysis of the remaining 6819 images revealed the main explanatory components for the object formation characteristics of outdoor social media images (Table 3). Four principal components with eigenvalues greater than 1 together explained 71.043% of the variance in the seven object categories inside urban parks. Loadings of the seven object categories inside urban parks on the four principal components are exhibited in Table 4. With higher loadings for tree and impervious surface, PC1 represented a mixture of natural and artificial objects. PC2 was dominated by artificial objects, including human, impervious surface, and facility. PC3 and PC4 were both dominated by natural objects. Specifically, PC3 had higher loadings for tree and grass, and PC4 had a higher loading for flower.

Based on the loadings of the four principal components shown above,

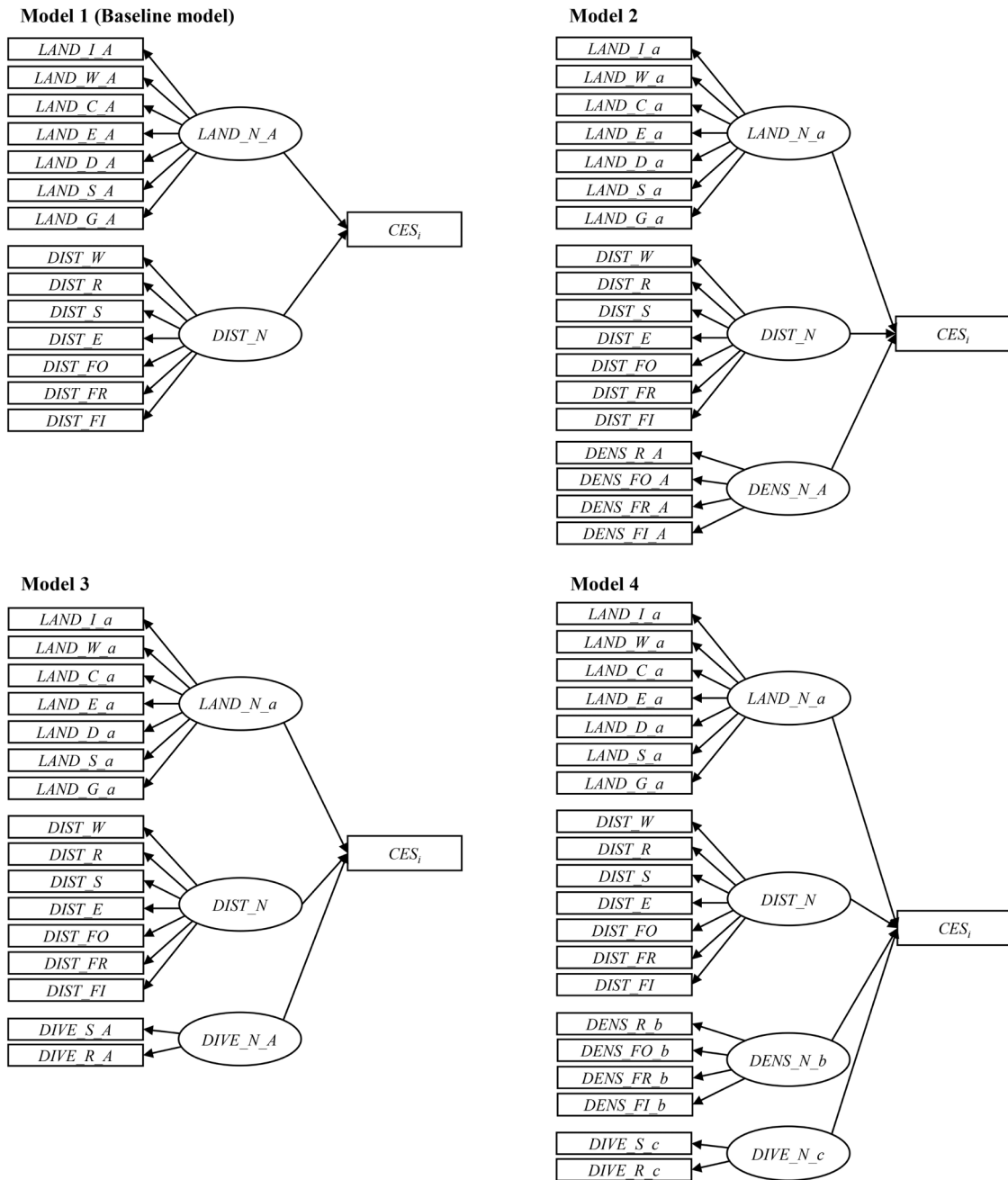


Fig. 4. Hypothetical models based on SEM.

Table 3
Initial eigenvalues of the principal component analysis.

Principal component	Eigenvalue	% of variance	Cumulative % of variance
1	1.556	22.223	22.223
2	1.238	17.687	39.910
3	1.038	14.828	54.738
4	1.001	14.306	71.043
5	0.938	13.399	82.443
6	0.629	8.981	91.423
7	0.600	8.577	100.000

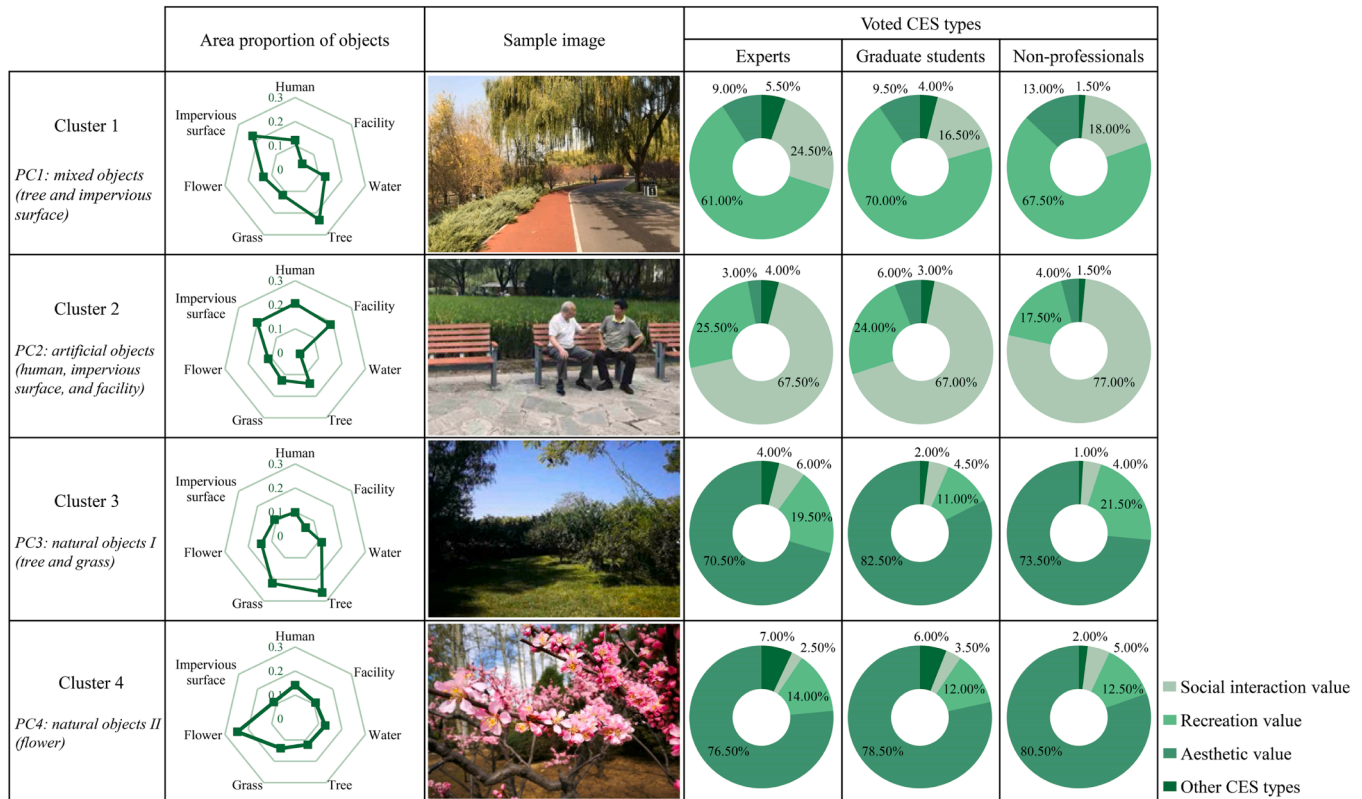
the *k*-means cluster analysis identified four clusters of outdoor social media images. The radar diagrams in Fig. 5 show the object formation characteristics of the four outdoor social media image clusters; the CES types voted by experts, graduate students, and non-professionals showed

similar trends for the four clusters. The images in Cluster 1 showed places where people could engage in physical activities and sightseeing, were voted to represent recreation value (63.00% of experts, 71.00% of graduate students, and 66.50% of non-professionals). The images in Cluster 2 exhibited places that could support group communication, and were voted as representing social interaction value (67.50% of experts, 67.00% of graduate students, and 77.00% of non-professionals). Images in Clusters 3 and 4 were made up mostly of natural scenery and close-up shots of plants, were both voted to represent the aesthetic value of urban parks (70.50% and 76.50% of experts, 82.50% and 78.50% of graduate students, and 73.50% and 80.50% of non-professionals). According to the questionnaire results, clusters of outdoor social media images were assigned to represent specific CES types. Statistically, 42.61% of the outdoor social media images of the surveyed parks represented recreation value, 33.50% represented aesthetic value, and only 23.89% represented social interaction value.

Table 4

Loadings of the principal components.

Principal components	Object categories						
	Human	Facility	Water	Tree	Grass	Flower	Impervious surface
PC 1: mixed objects	-0.111	-0.719	-0.068	0.686	-0.146	-0.016	0.639
PC 2: artificial objects	0.569	0.434	-0.836	0.085	-0.027	-0.124	0.542
PC 3: natural objects I	-0.331	-0.618	-0.231	0.766	0.484	-0.018	-0.253
PC 4: natural objects II	0.016	-0.216	-0.075	-0.125	-0.004	0.766	-0.188

**Fig. 5.** Clusters of outdoor social media images and voted CES types.

3.1.2. Maps of CES in surveyed urban parks

Based on the clustering analysis of the segmented objects and CES types voting, the spatial distributions of recreation value, aesthetic value, and social interaction value in the surveyed urban parks were mapped (Fig. 6). The Kruskal-Wallis one-way ANOVA test confirmed that the three maps were significantly different from each other ($p < 0.01$). The distributions of the three CES types showed that they did not occur randomly, but exhibited spatial clustering. The recreation value was more tightly linked to the locations of park roads, whereas the aesthetic and social interaction values were more strongly related to locations of facilities and specific land cover types. This was statistically confirmed by spatial autocorrelation tests; the global Moran's I index was 0.977 for recreation value ($Z = 304.656$, $p < 0.001$), 0.972 for aesthetic value ($Z = 304.441$, $p < 0.001$), and 0.967 for social interaction value ($Z = 304.347$, $p < 0.001$). The clustering patterns of the three CES types indicated that distributions of CES corresponded with specific urban park features, making it possible to explore their association and enhance CES through design and management practices.

3.2. Urban park features for explaining CES

3.2.1. Four aspects of urban park features

Results of the SEM analyses are shown in the [Supplementary](#)

[Materials](#) (Tables S6–S9). The four aspects of urban park features all had a significant impact on the three CES types ($p < 0.01$). Table 5 shows the goodness of fit for the models at each service radius. Generally, the model results showed that the goodness-of-fit of Model 4 for the three CES types was uniformly better than that of the other models (R -squared for recreation value, aesthetic value, and social interaction value was 0.469, 0.499, and 0.561, respectively) and confirmed the first hypothesis that a combination of the four aspects of urban park features performed the best in explaining CES. The goodness of fit of Models 2 and 3 was also better than the baseline model, suggesting that in addition to the extensively considered urban park features of land cover types and distance to elements, density of elements and landscape diversity also significantly contribute to explaining CES.

The estimated results of Models 1–3 confirmed the second hypothesis that urban park features within different services radii affect CES to varying extents and identified the most significant service radius for each urban park feature in explaining CES. For recreation value, the combination of land cover proportion within a 100 m radius and the density of elements and landscape diversity within a 50 m radius resulted in the most significant model ($a = 100$, $b = 50$, $c = 50$). For aesthetic value, the best model involved the combination of land cover proportion within a 100 m radius, density of elements within a 30 m radius, and landscape diversity within a 50 m radius ($a = 100$, $b = 30$, c



Fig. 6. Maps of the three CES types.

Table 5
Goodness of fit of the structural equation models.

Models	Service radius	CES _i		
		Recreation value	Aesthetic value	Social interaction value
Model 1	A= 10	0.344 *	0.397 *	0.412 **
	A= 30	0.361 **	0.417 ***	0.441 ***
	A= 50	0.372 ***	0.431 ***	0.494 ***
	A= 100	0.396 ***	0.442 ***	0.463 ***
	A= 10	0.399 ***	0.444 ***	0.497 ***
Model 2	A= 30	0.461 ***	0.486 ***	0.559 ***
	A= 50	0.464 ***	0.484 ***	0.553 ***
	A= 100	0.433 ***	0.465 ***	0.542 ***
	A= 10	0.398 ***	0.447 ***	0.497 ***
	A= 30	0.405 ***	0.457 ***	0.501 ***
Model 3	A= 50	0.407 ***	0.459 ***	0.502 ***
	A= 100	0.403 ***	0.454 ***	0.499 ***
	/	0.496 ***	0.499 ***	0.561 ***
	4			

Note: Numbers in the table are the *R*-squared of the models. Bold numbers are the most significant estimates of each model. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

= 50). Finally, for social interaction value, the best combination was land cover proportion and landscape diversity within a 50 m radius and density of elements within a 30 m radius ($a = 50$, $b = 30$, $c = 50$).

3.2.2. Design-related indicators

Tables 6–8 show the results of multiple stepwise regression analysis

Table 6
Multiple stepwise regression of recreation value.

	β	Std. Error	t value	Robust_Pr	VIF
Intercept	0.233	0.010	24.178	< 0.001	–
LAND_C_100	0.320	0.014	22.675	< 0.001	1.691
LAND_I_100	0.015	0.006	2.505	0.008	1.389
DIST_E	0.204	0.009	21.639	< 0.001	1.818
DIST_R	-0.421	0.015	-27.373	< 0.001	2.071
DIST_W	-0.095	0.010	-9.834	< 0.001	1.615
DIST_G	-0.070	0.013	-5.268	< 0.001	2.102
DIST_FO	-0.384	0.012	-6.618	< 0.001	2.770
DIST_FI	-0.386	0.009	-21.820	< 0.001	2.046
DIVE_R_50	0.158	0.008	7.641	< 0.001	1.165
DENS_R_50	0.375	0.011	58.851	< 0.001	3.049
DENS_FO_50	0.182	0.041	3.009	< 0.001	1.056
DENS_FI_50	0.472	0.027	-4.444	< 0.001	1.180
DENS_FR_50	0.124	0.015	12.175	< 0.001	2.038

of the various CES types. Table 6 shows that the distances to interactive facilities ($\beta = -0.486$) and ornamental facilities ($\beta = -0.484$) contributed most to explaining recreation value, indicating that places close to interactive facilities and ornamental facilities may have greater potential to supply recreation value. Road density within a 50 m radius ($\beta = 0.375$) and proportion of canopy-shaded ground within a 100 m radius ($\beta = 0.320$) are also relatively important to recreation value. As shown in Table 7, the proportion of canopy-shaded ground within a 100 m radius ($\beta = 0.743$) contributed most to aesthetic value, followed by road density within a 30 m radius ($\beta = 0.386$) and distance to ornamental facilities ($\beta = -0.337$); landscape density within a 50 m radius was also

Table 7
Multiple stepwise regression of aesthetic value.

	β	Std. Error	t value	Robust_Pr	VIF
Intercept	0.292	0.011	26.551	< 0.001	–
LAND_C_100	0.393	0.015	48.077	< 0.001	2.176
LAND_D_100	0.153	0.008	14.517	< 0.001	1.491
LAND_I_100	-0.126	0.008	-16.736	< 0.001	1.547
LAND_W_100	0.119	0.009	2.157	0.002	2.284
DIST_R	-0.255	0.016	-15.907	< 0.001	2.450
DIST_W	-0.269	0.010	16.944	< 0.001	1.841
DIST_G	0.102	0.013	7.829	< 0.001	2.191
DIST_FR	-0.085	0.016	-30.767	< 0.001	3.093
DIST_FO	-0.337	0.011	-31.337	< 0.001	2.589
DIST_FI	-0.186	0.008	-11.357	< 0.001	1.732
DIVE_R_50	0.230	0.011	20.175	< 0.001	3.166
DIVE_S_50	0.069	0.010	6.864	< 0.001	3.292
DENS_R_30	0.253	0.011	35.484	< 0.001	1.925
DENS_FO_30	0.486	0.017	9.243	< 0.001	1.301
DENS_FI_30	0.066	0.021	-3.197	< 0.001	1.731

Fit of the model: $R^2m = 0.544$, $R^2a = 0.543$, F -statistic = 2861.787, $p < 0.001$.

Table 8
Multiple stepwise regression of social interaction value.

	β	Std. Error	t value	Robust_Pr	VIF
Intercept	0.228	0.010	23.060	< 0.001	–
LAND_C_50	0.334	0.013	32.518	< 0.001	1.706
LAND_D_50	0.181	0.009	20.333	< 0.001	1.994
LAND_G_50	0.324	0.014	23.671	< 0.001	1.283
LAND_I_50	-0.114	0.007	-16.733	< 0.001	1.990
DIST_E	0.047	0.009	5.274	< 0.001	1.828
DIST_R	0.051	0.016	3.245	< 0.001	2.457
DIST_W	0.191	0.010	17.549	< 0.001	1.839
DIST_G	-0.108	0.012	-8.709	< 0.001	2.072
DIST_FR	-0.473	0.016	-29.420	< 0.001	3.354
DIST_FO	-0.242	0.011	-22.410	< 0.001	2.730
DIST_FI	-0.072	0.008	-9.065	< 0.001	2.036
DIVE_R_50	0.170	0.008	8.686	< 0.001	1.656
DIVE_S_50	0.141	0.014	68.890	< 0.001	2.546
DENS_FR_30	0.494	0.017	25.157	< 0.001	1.187
DENS_FO_30	0.149	0.036	13.727	< 0.001	1.100

Fit of the model: $R^2m = 0.567$, $R^2a = 0.567$, F -statistic = 3599.263, $p < 0.001$. Fit of the model: $R^2m = 0.525$, $R^2a = 0.524$, F -statistic = 2776.301, $p < 0.001$.

important ($\beta = 0.230$). This emphasized the importance of ornamental facilities and diverse landscape variations to aesthetic value. For social interaction value (Table 8), the distance to resting facilities ($\beta = -0.473$) and the proportion of canopy-shaded ground within a 50 m radius ($\beta = 0.434$) contributed most to explain its supply, followed by proportion of grass within a 50 m radius ($\beta = 0.324$), indicating that people tend to socialize in shaded places where they can sit.

3.3. Urban park design strategies for enhancing CES

As shown in Table 9, this study proposed a summary of evidence-based strategies for landscape designers and managers according to the multiple stepwise regression analysis results, which will provide references for landscape site design in urban parks to enhance specific CES types. Specifically, to enhance recreation value, the landscape site should be located near park roads and entrances to increase its accessibility, ensuring a large proportion of tree canopy-shaded ground within a 100 m radius was suggested to optimize the microclimate of active spaces, and diverse land cover types with various interactive facilities should be arranged within a 50 m radius. For a landscape site with relatively lower aesthetic value, adaptive enhancements should simultaneously consider designing extensive tree canopy-shaded ground near water areas, creating near-natural scenery and visual effects with combinations of different vegetation types, and arranging various ornamental facilities within a 30 m radius. To enhance social interaction value, the landscape site should be located far away from water and

main roads to where may be popular for gathering, a large proportion of tree canopy-shaded ground was also needed within a 50 m radius to create spaces with a comfortable microclimate, and resting facilities coordinated with proper ornamental facilities are suggested to be arranged within a 30 m radius. In practice, these strategies still need to be properly integrated and adjusted to adapt to the current status of landscape sites.

4. Discussion

4.1. The potential of outdoor social media images in identifying CES

This study addressed the first research question with a computer vision approach and a questionnaire survey. Three CES types in urban parks were identified from geo-tagged outdoor social media images, including recreation, aesthetic, and social interaction values. Based on automated content identification of outdoor social media images, this study further attempted to integrate and distinguish the perspectives of experts, graduate students, and non-professionals in identifying different CES types. The similarity of the voting results indicated that people tend to share similar perceptions of the function and value of urban park elements, regardless of whether they have ecosystem services-related knowledge. Some previous studies have designated CES types to be related to specific objects based on researchers' experience (Richards and Friess, 2015; Thiagarajah et al., 2015; Retka et al., 2019; Cardoso et al., 2022). However, this view may ignore the synergies between different objects and be limited by expertise bias because earlier findings confirmed that experts and laypersons understand CES differently (La Rosa et al., 2016; Riechers et al., 2017). This study was a rewarding attempt to identify CES types based on social media data from the public participation perspective. Although the voting results of experts, graduate students, and non-professionals in this study exhibited no obvious differences, Fig. 5 indicates that experts and graduate students identified more images as "other CES types", especially experts. This was consistent with previous findings that confirmed that experts tend to value a more balanced mixture of benefits (Riechers et al., 2017).

The three identified CES types in this study were also the three most widespread reasons for urban park use in China (Wang et al., 2021) and the most focused CES types in previous literature (La Rosa et al., 2016). Based on the relatively precise positioning of outdoor social media images, the intensity value of different CES types in the surveyed urban parks were mapped by taking the density of images as an indicator of public interest in urban park sites (Richards and Friess, 2015). Results for the mapped CES revealed aggregated patterns, and was consistent with earlier findings (Gerstenberg et al., 2020; Havinga et al., 2020; Tian et al., 2021). Compared to previous literature based on social media images, only three of the ten given CES types were identified in this study. This can be explained that earlier studies were generally conducted on larger scales, which contain more diverse land cover types (Yoshimura and Hiura, 2017; Tieskens et al., 2018; Depietri et al., 2021; Fox et al., 2021), whereas the small-scale urban parks analyzed in this study may be limited to providing all-sided CES types. Moreover, CES types such as sense of place and inspirational value have more complex relationships with the various elements of urban parks and may be affected by specific social and historical contexts (Hernández-Morcillo et al., 2013; Milcu et al., 2013), may be more difficult to capture by outdoor social media images. Although these CES types were not identified in this study, it could be inferred that some of the images in urban parks were responses to an inspirational environment and a sense of belonging (Richards and Friess, 2015).

4.2. Urban park features and design-related indicators that affect CES

By incorporating comprehensive urban park features, this study addressed the second research question through a series of statistical analyses, i.e., how do urban park features contribute to explaining CES.

Table 9
Strategies for enhancing CES in a landscape site.

CES type	Strategies			
Recreation value				
	Enlarging canopy-shaded ground within 100 m.	Close to ornamental and interactive facilities.	Improving landscape diversity within 50 m.	Enriching interactive facilities within 50 m.
Aesthetic value				
	More tree canopies and water within 100 m.	Close to ornamental facilities and water.	Improving landscape diversity within 50 m.	Enriching ornamental facilities within 30 m.
Social interaction value				
	More tree canopies and grass within 50 m.	Close to resting facilities.	Improving landscape diversity within 50 m.	Enriching resting facilities within 30 m.

The SEM analyses proved that a combination of comprehensive urban park features within a specific service radius performed the best in the hypothesis model estimates. Although previous findings have identified and suggested attaching importance to various landscape features to enhance CES and outdoor activities (Pietilä and Kangas, 2015; Zhai and Baran, 2017; Gerstenberg et al., 2020), the ranges within which these landscape features should be arranged remain unknown. Results of this study suggest that compared to recreation value and aesthetic value, social interaction value in urban parks is more significantly affected by urban park features within a relatively small service radius. One explanation for this may be that when socializing, people are likely to care about relatively quiet immediate surroundings and pay more attention to their close interactions (Rivera et al., 2021; Veitch et al., 2022). In contrast, people engaged in recreational activities or enjoying scenery tend to appreciate attractive places which could encourage visitors to approach (Zhai and Baran, 2017; Talal and Santelmann, 2021). These findings could be valuable for informing design practice on more detailed scales because arranging various landscape features within specific sites is one of the crucial tasks in urban park design (Starke and Simonds, 2013).

The results of the multiple regression analyses further identified the importance of specific design-related indicators in affecting different CES types. Previous findings on large scales emphasized that the occurrence of natural-related social media images was best predicted by the distance to outdoor attractions and roads (Sherrouse et al., 2011; Richards and Tunçer, 2018). In this study, the density of specific facilities was proved to be more important than distance-related indicators for all the three CES types. One explanation of this finding may be that park visitors would rather visit places or sights with a variety of facilities than those with only one (Veitch et al., 2022). Distance to park roads was also found to be significantly related to recreation and aesthetic values, but to have no significant relationship with social interaction value. This indicates that landscape sites near park roads were more attractive to people, whereas people socializing may prefer places away from main roads. It was also confirmed that the proportion of tree canopy-shaded ground was important for all the three CES types, which was consistent with previous studies that revealed the importance of tree canopy-shaded ground in providing a cooling effect in urban parks

proposed for leisure and recreational activities (Colter et al., 2019; Xu et al., 2019; Yu et al., 2020). The richness index of land cover types also proved to be important for all the three CES types because diverse land cover types contribute to greater species richness (Tscharntke et al., 2012) and to a greater visual preference effect due to increased scenic complexity and variety (Dronova, 2017; Ha et al., 2022), which has proved to be positive for CES preference or perception (Linde-mann-Matthies et al., 2010).

4.3. Implications for urban park design practices to enhance CES

Based on the results of multiple regression analyses, this study further addressed the third research question, i.e., how to enhance CES through urban park design practices. Design-related indicators that affect different CES types were identified within specific service radii, and practical urban park design strategies were proposed for enhancing CES in landscape sites. Maps of the different CES types could provide urban park designers with spatial explicit information about where optimal planting, paving, and facility layout solutions should be implemented. To further explain the third research question based on the research results, we proposed three adaptive landscape site design cases by taking Haidian Park (HD) as an example.

As shown in Fig. 7, three sites were selected for enhancing recreation value, aesthetic value, and social interaction value respectively. With simple vegetation and few facilities, Site 1 was optimized to promote recreation value. More canopy-shaded ground was created, with accessible physical facilities and various vegetation types. Site 2 was re-designed to enhance aesthetic value by arranging ornamental facilities in the middle of the grass area and designing areas of various vegetation types to nurture near-natural scenery and visual effect. In Site 3, more private spaces were created with various vegetation types and equipped with various resting facilities to provide quiet and convenient social interaction spaces to enhance social interaction value.

4.4. Limitations and research prospects

Based on the computer vision approach and the questionnaire survey, this study identified three CES types in urban parks with outdoor

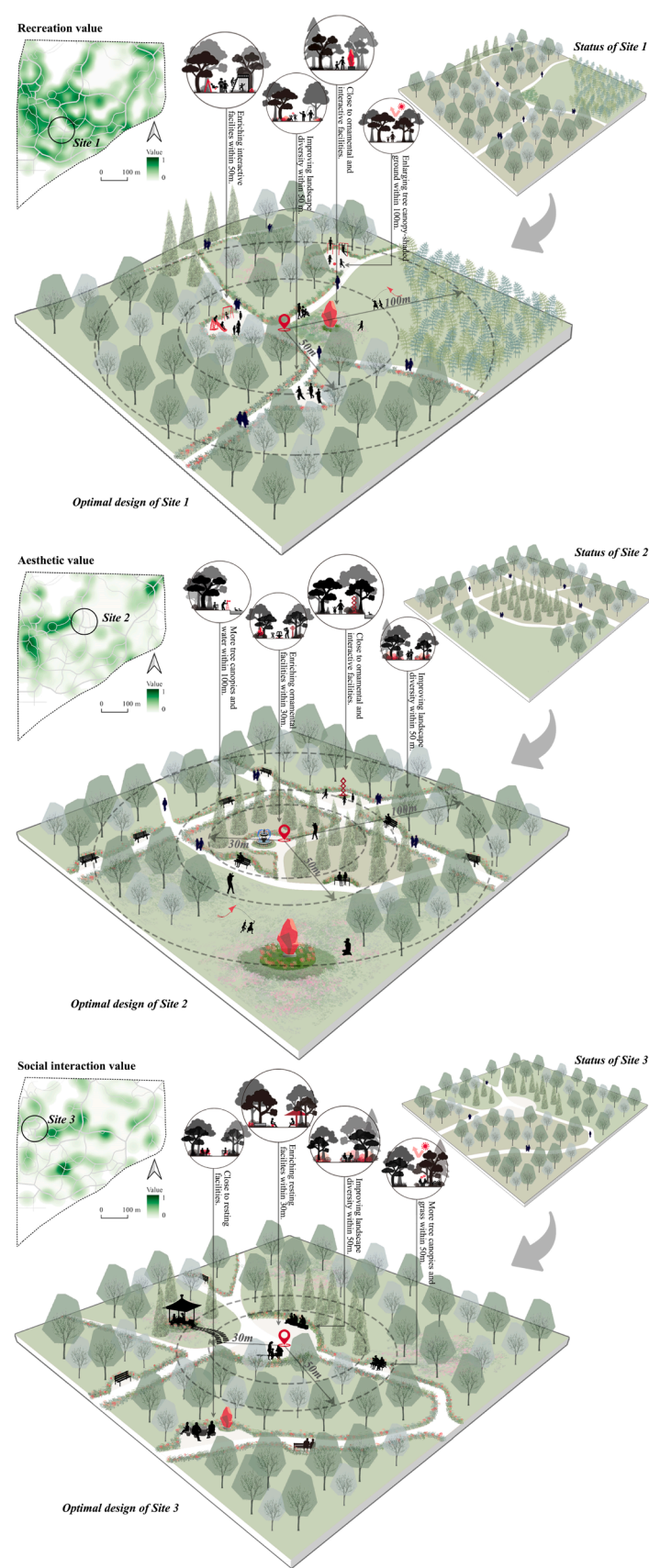


Fig. 7. Optimal designs for enhancing CES.

social media images. However, it can be inferred that users of outdoor social media are mainly outdoor enthusiasts, who are a group that may prefer to be physically active in urban parks rather than enjoying scenery or communicating with others. This may result in deviations from the identified CES types in the surveyed urban parks. It is unclear whether the result that more than 40% of the outdoor social media images were voted as representing recreation value can be attributed to the deviation because recreation value has also been widely confirmed to be the most important and appreciated CES type in urban parks (Liu et al., 2020; Wang et al., 2021).

This study has summarized a series of strategies for specifically enhancing different CES types, which can be adaptively used in landscape site design cases. But these strategies are still undefined for thresholds. For example, it has been identified that high density of park roads can contribute to enhancing both recreation and aesthetic value. However, the extent to which this promotion can be effective is unclear, given that extremely high road density is unrealistic in urban parks. The same problem also occurs for the feature of distance to elements; although nearby ornamental facilities and interactive facilities make similar prominent contributions to recreation value, it is impractical to arrange all these facilities close to park visitors. Thresholds for these urban park features and their trade-offs should be further researched to provide more exact information for urban park design practices. More aspects of urban park features and indicators related to urban park design can also be further assessed and considered in expanded analyses.

Finally, many recent studies have focused on the demand for CES (Hegetschweiler et al., 2017; Bing et al., 2021; Zhao et al., 2022) and have emphasized assessment of the amount of CES required or desired by the society (Nedkov and Burkhard, 2012; Villamagna et al., 2013). Although this study identified places where CES enhancement is needed and proposed three optimal design cases, the selection of case sites was based on the current intensity of CES. Demands for CES from park visitors should also be further considered in decision-making. However, assessment of CES demand at present mostly relies on interviews, questionnaire surveys, and PPGIS approaches with large time and labor costs (Bing et al., 2021). This study confirmed that CES can be identified at a more detailed scale based on outdoor social media data. More crowdsourced data could be further considered to spatially measure park visitors' requirement and desire for CES and provide more effective information for urban park design practices.

5. Conclusion

Evidence-based design strategies are needed for managers and designers to manage urban parks to promote CES and related human health and well-being outcomes. This study attempted to bridge the gap between CES assessment research and urban park design practices with a practice-oriented workflow. In six urban parks in Beijing, three CES types, including recreation value, aesthetic value, and social interaction values, were identified on a more detailed scale based on outdoor social media images processed with a computer vision approach and a questionnaire survey. The statistical analyses proved that a combination of the four aspects of urban park features within specific ranges was most effective in explaining different CES types. Compare to recreation value and aesthetic value, social interaction value in urban parks was proved to be more significantly explained by urban park features within relatively small service radii. The importance of design-related indicators in affecting different CES types was also identified. Density of different facilities was proved to be most important in affecting the three CES types in surveyed urban parks. Tree canopy-shaded ground and richness of land cover types were also proved positive for the three CES types, whereas indicators about park roads affected only recreation and aesthetic values. Landscape site design strategies and three adaptive design cases for enhancing CES in urban parks were proposed based on the quantified associations between urban park features and CES. It was concluded that based on the practice-oriented workflow, the results of

this study could be adaptively used to inform urban park design practices for enhancing CES. Further research into exploring additional aspects of urban park features and thresholds for design-related indicators that affect CES could provide more exact information to evidence-based design practices for sustainable cities, human health, and well-being.

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CRedit authorship contribution statement

Yanan Wang: Methodology, Investigation, Writing – original draft, Formal analysis, Visualization. **Xiaoxiao Shi:** Investigation. **Kejun Cheng:** Visualization. **Jingke Zhang:** Visualization. **Qing Chang:** Conceptualization, Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ufug.2022.127713](https://doi.org/10.1016/j.ufug.2022.127713).

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